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EXPERIMENT 10

Exp No:10 Experiment to understand array function in Data science.

AIM:
To understand array function in Data science.

DESCRIPTION:
Understand array function using Numpy library

CODE:

```
import numpy as np

array = np.array([72, 18, 45, 91, 36, 64, 27, 83, 12])

print("array:")

print(array)

print("\nsquare root:")

print(np.sqrt(array))

print("\nnumber of dimensions:")

print(array.ndim)

new_array = array.reshape(3,3)

print("\nreshaped array:")

print(new_array)

print("\nnumber of dimensions:")

print(new_array.ndim)

print("\nravel:")

print(new_array.ravel())
```

```
newm = new_array.reshape(3,3)
print("\nnewm:")
print(newm)
print("\nnewm[2,1:3]:")
print(newm[2,1:3])
print("\nnewm[1:2,1:3]:")
print(newm[1:2,1:3])
print("\nnew_array[0:3,0:0]:")
print(new_array[0:3,0:0])
print("\nnew_array[0:2,0:1]:")
print(new_array[0:2,0:1])
print("\nnew_array[0:3,0:1]:")
print(new_array[0:3,0:1])
print("\nnew_array[1:3]:")
print(new_array[1:3])
```

OUTPUT:

```
array:
[72 18 45 91 36 64 27 83 12]
square root:
[8.48528137 4.24264069 6.70820393 9.53939201 6.         8.
 5.19615242 9.11043358 3.46410162]
number of dimensions:
1
reshaped array:
[[72 18 45]
```

[91 36 64]

[27 83 12]]

number of dimensions:

2

ravel:

[72 18 45 91 36 64 27 83 12]

newm:

[[72 18 45]

[91 36 64]

[27 83 12]]

newm[2,1:3]:

[83 12]

newm[1:2,1:3]:

[[36 64]]

new_array[0:3,0:0]:

[]

new_array[0:2,0:1]:

[[72]

[91]]

new_array[0:3,0:1]:

[[72]

[91]

[27]]

new_array[1:3]:

[[91 36 64]

[27 83 12]]

RESULT:

The array operations were successfully performed using NumPy, including reshaping, indexing, and slicing.